

Cover Letter - TMLR Submission Draft

To: TMLR Action Editor

From: Zewen Liu (■■■■)

Date: 2026-08-02

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Re: Single-paper submission

Dear Editor,

We submit the manuscript "*AGI Kit: An End-to-End Self-Improving Tool-Use Pipeline on Consumer Hardware - Empirical Observations*" for consideration at TMLR. The repository URL will be inserted before submission.

The manuscript reports a four-layer tool-use pipeline comprising per-step reflection, semantic strategy memory, continual learning with an A/B safety gate, and bounded schema mutation. The system runs on CPU-only consumer hardware with a Qwen3-1.7B primary model and a Qwen3-0.6B scorer.

The paper's main contribution is a transparent empirical system report, including negative results and identified confounds:

- A five-configuration synthetic ablation is saturated at 77.8% JSON final-emission for every configuration and does not identify a component effect.
- On a 20-task arithmetic evaluation, the bare model scores 5.0% at three steps and 35.0% at six steps, while the full configuration records 100.0% structural completion but 95.0% correctness after an independent gold recheck. We report the 60-point matched-step gap as a configuration difference, not a pure layer effect, because prompt structure and control flow also differ.
- A new prompt-, tool-, budget-, and seed-matched 20-task check makes an L1 tool-evidence correction executable: Static scores 19/20 and L1 scores 20/20. The single discordant pair has exact two-sided McNemar $p=1.0$, so we describe it as an auditable correction instance rather than a statistically supported general gain.
- Re-evaluation of 138 historical episodes separates structural final emission from answer correctness; correctness is 77.6% on the 85-episode arithmetic subset.
- The A/B safety gate passes 12 deterministic boundary tests. The production SchemaMutator rejects 18 invalid field/value changes and accepts 12 valid in-range controls in a direct policy test.
- Prompt-template comparisons include both negative and exploratory positive results. They are not presented as architecture-level comparisons against full ReAct, Reflexion, Voyager, or MetaGPT implementations.

We explicitly invalidate the inferential interpretation of an earlier three-run summary: the driver produced 16 rather than 15 episodes per run, did not control every stochastic seed, and later reused output directories. The revised manuscript reports those values descriptively and makes no confidence-interval, power, or superiority claim.

The repository contains source code, task data, experiment scripts, figures, and machine-readable result summaries. The manuscript also states the principal limitations: no canonical end-to-end GAIA2 harness, no architecture-level baseline comparison, incomplete seed control for the historical repeated runs, mock retraining in the headline continual-learning experiment, and no real-world deployment.

We have no conflicts of interest to declare.

Sincerely,

Zewen Liu